

THYBOROEN, Denmark

WMO: 060520

Lat: 56.7067N

Lon: 8.2150E

Elev: 13

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.0	24.7	16.9	12.9	25.0	19.4	14.6	26.7	42.0	36.9	37.3	41.2	12.4	50	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	76.0	65.2	72.4	63.9	69.6	62.9	66.7	73.3	65.1	70.7	63.7	68.5	11.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	89.7	69.7	62.8	85.5	68.2	61.5	81.6	66.8	31.3	73.6	30.0	70.8	29.0	68.7	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
37.8	33.1	29.7	DB	19.9	82.1	3.8	4.3	17.1	85.2	14.9	87.7	12.7	90.1	9.9	93.2	
			WB	18.9	69.1	3.7	1.8	16.2	70.3	14.1	71.4	12.0	72.3	9.4	73.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	36.5	35.9	38.7	45.5	52.3	57.5	62.6	63.4	58.6	51.4	44.5	39.4
	DBStd	10.89	5.91	5.64	4.77	4.56	5.07	4.29	4.63	3.79	3.65	4.48	5.20	6.30
	HDD50	1883	417	396	351	151	29	0	0	0	0	37	173	329
	HDD65	5933	882	816	816	585	397	229	105	79	194	420	616	794
	CDD50	1496	0	0	0	16	98	226	392	414	259	82	8	1
	CDD65	71	0	0	0	0	1	6	32	29	3	0	0	0
	CDH74	194	0	0	0	0	5	17	109	59	3	0	0	1
	CDH80	19	0	0	0	0	0	0	15	4	0	0	0	0
(14) Wind	WSAvg	15.0	16.1	15.3	14.7	13.4	13.8	14.6	13.7	14.2	15.6	15.9	16.3	16.3
(15) Precipitation	PrecAvg	31.9	2.7	1.7	2.1	1.6	1.9	2.0	2.3	3.0	3.4	3.9	3.9	3.2
	PrecMax	40.8	5.9	4.8	5.1	3.3	9.1	4.9	5.9	6.9	7.8	9.4	8.0	8.1
	PrecMin	23.4	0.4	0.2	0.3	0.0	0.3	0.5	0.3	0.0	0.7	0.9	1.4	0.6
	PrecStd	4.8	1.3	1.1	1.2	0.9	1.6	1.2	1.3	1.8	1.8	2.1	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	46.6	52.4	63.7	73.4	76.3	82.1	79.8	72.6	63.1	56.8	50.9
		MCWB	46.1	44.1	46.7	53.0	60.7	63.4	66.9	67.4	63.8	58.4	54.8	48.9
	2%	DB	46.4	45.0	47.3	58.8	68.0	71.3	77.2	75.0	68.2	60.1	54.1	49.1
		MCWB	44.2	43.2	44.0	51.4	58.1	61.2	65.8	65.4	62.0	56.8	51.9	47.1
	5%	DB	45.0	44.0	45.4	55.4	63.5	66.9	72.7	71.4	65.3	58.6	52.6	47.9
		MCWB	43.1	42.4	42.9	49.7	56.4	60.0	64.5	64.6	60.5	55.7	50.3	46.0
	10%	DB	43.9	42.9	44.1	52.2	59.4	63.6	69.4	68.8	63.5	57.2	51.1	46.8
		MCWB	42.1	41.4	42.0	47.9	54.2	58.6	63.6	63.2	59.5	54.6	48.7	44.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	44.9	46.9	54.8	63.0	65.1	69.0	68.6	65.1	59.5	55.0	49.2
		MCDB	47.7	45.8	50.3	60.7	70.8	72.3	77.9	75.8	69.9	62.1	56.3	50.2
	2%	WB	44.8	43.7	44.8	52.2	59.5	62.7	67.0	66.8	63.0	57.6	52.4	47.5
		MCDB	45.9	44.7	46.6	57.3	65.8	69.6	74.0	73.1	66.7	59.4	53.7	48.6
	5%	WB	43.6	42.6	43.5	50.2	56.9	60.8	65.4	65.2	61.6	56.3	50.8	46.4
		MCDB	44.7	43.7	45.0	54.3	62.3	66.0	71.3	70.5	64.7	58.1	52.2	47.6
	10%	WB	42.5	41.6	42.3	48.3	54.5	58.9	63.8	63.8	60.2	55.0	49.1	45.2
		MCDB	43.7	42.7	43.7	51.7	59.0	62.9	68.9	68.2	63.0	56.9	50.7	46.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	6.0	8.1	8.3	7.4	7.6	7.3	6.4	5.7	5.1	5.2
		MCDBR	4.7	4.5	7.5	14.3	15.6	14.0	14.4	11.9	9.6	6.6	4.9	4.8
		MCWBR	5.0	4.2	5.4	8.3	8.8	7.6	6.5	6.0	6.0	5.4	5.5	5.2
	5% WB	MCDBR	4.6	4.3	6.7	12.7	14.0	12.6	12.3	10.3	8.7	6.3	4.9	4.4
		MCWBR	5.1	4.2	5.2	8.1	8.7	7.9	6.4	6.0	6.1	5.7	5.6	5.3
(40) Clear-Sky Solar Irradiance	taub	0.321	0.341	0.358	0.392	0.381	0.376	0.389	0.393	0.373	0.352	0.329	0.304	
	taud	2.215	2.279	2.324	2.280	2.377	2.416	2.396	2.388	2.427	2.390	2.276	2.201	
	Ebn at Noon	172	223	252	260	272	275	268	258	246	218	169	145	
	Edh at Noon	17	24	30	36	35	34	34	32	27	22	16	14	
(44) All-Sky Solar Radiation	RadAvg	133	318	755	1274	1695	1834	1692	1332	895	445	169	93	
	RadStd	25	50	69	145	161	150	173	102	89	52	29	11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	-229	-304	N/A	N/A

Nomenclature: See separate page